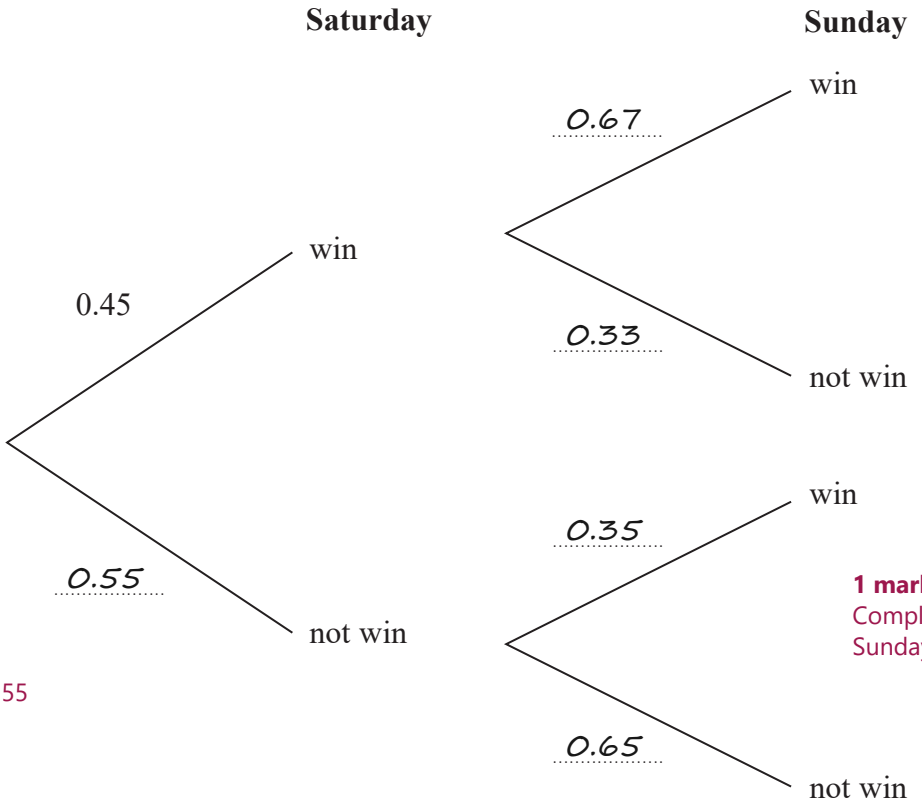
15 A darts team is going to play a match on Saturday and on Sunday.
The probability that the team will win on Saturday is 0.45

If they win on Saturday, the probability that they will win on Sunday is 0.67
If they do **not** win on Saturday, the probability that they will win on Sunday is 0.35

(a) Complete the probability tree diagram.



1 mark
1 − 0.45 = 0.55

1 mark
Complete all four Sunday probabilities

(2)

(b) Find the probability that the team will win exactly one of the two matches.

Exactly one win: win then not win, OR not win then win.

$0.45 \times 0.33 = 0.1485$

1 mark
Multiply along a path

$0.55 \times 0.35 = 0.1925$

1 mark
Add the two required paths

$0.1485 + 0.1925 = 0.341$

1 mark
Correct probability

0.341

(3)

(Total for Question 15 is 5 marks)